Distinguished Lecture Review

Talk title	Computational Complexity and Explanations in Physics
Speaker	Prof. Scott Aaronson (University of Texas at Austin)
Quarter/year talk was given	Winter 2026 (delivered Thursday, January 8, 2026, 3:30–4:30 PM)
Talk URL/link	https://www.youtube.com/watch?v=GcbLL4C3VwY
Talk was a	Distinguished Lecture
Lecture notes	Lecture Notes
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1. Heart of the Talk & Great Ideas

Prof. Aaronson's central claim, as I understood it, is that computational complexity is not merely an engineering constraint on what we can compute, but a *physical* constraint on what can happen — and therefore a tool for explaining physics itself. The talk sweeps from Euclid's GCD algorithm through Turing's formalization of computability in the 1930s, out to the LHC computing clusters that analyze particle collisions and the AI infrastructure now absorbing a significant fraction of the world economy. The historical arc set up the question for me: if time, memory, randomness, and quantumness are resources the universe has to actually spend, then maybe complexity classes aren't abstractions — they're bookkeeping.

One idea I kept turning over afterward was the reframing of quantum mechanics as **"probability theory with minus signs."** Many popular introductions to quantum mechanics lean on the mystique of superposition; Prof. Aaronson instead demystifies it by showing that if you let probability amplitudes go negative (or complex) and square at the end, destructive interference just drops out of the arithmetic. The double-slit example with $\alpha_1 = ½$, $\alpha_2 = -½$ giving $P = |\alpha|^2 = 0$ felt like the most accessible entry point I've come across, and — at least in my head — it re-situated the weirdness of quantum mechanics as a small mathematical generalization rather than a metaphysical leap.

Another part I found striking was his treatment of the **hype-versus-reality gap in quantum computing**. He does the arithmetic most popular accounts skip: Grover gives only a $\sqrt{N}$ speedup, not exponential; quantum error correction currently imposes roughly 10^6 overhead per logical qubit; so for Grover to beat classical, the problem has to be larger than about 10^{12} items. That single calculation changed how I hear the phrase "quantum advantage" — less a slogan, more a measurable threshold. His pointer to **Ewin Tang's dequantization** made the same point on the ML side: many quantum-ML wins can be matched classically. As a UW PMP student, this was a highlight.

The part I found most philosophically interesting was **David Deutsch's 1997 challenge**: when Shor's algorithm factors a number using computational resources $\sim 10^{500}$ beyond what the visible universe (10^{80} atoms) seems able to support, *where* is the computation happening? The question seems to force a choice — Many-Worlds supplies an answer ("in the other branches"), Copenhagen refuses the question, Bohmian mechanics posits a privileged real branch. Prof. Aaronson's framing — **"two worlds colliding, that's a quantum computer"** — stuck with me. He extends this with the Harlow–Hayden argument that decoding Hawking radiation would require a doubly exponential computation (~ 2^(10^{67})), which seems to dissolve the black-hole firewall paradox on purely complexity-theoretic grounds. That struck me as a new kind of physical argument: complexity isn't bounding what we can engineer, it's bounding what physical scenarios can even coherently exist.

Running through the whole talk was a third idea I'm still chewing on: physics enforces complexity-class boundaries. The Zeno-style "finite-time Goldbach decider" fails not for a mathematical reason but a thermodynamic one — halve your clock cycle enough times and the energy density exceeds the Schwarzschild limit; your CPU literally becomes a black hole. Similarly, if P = NP, you could appear to violate the Second Law of Thermodynamics by building a Maxwell's demon. The unifying thesis — that there is essentially a **conservation law for hardness**, in which time, energy, space, information, and amplitude structure trade off against each other but never shrink in product — is the takeaway I'm most likely to carry forward.

A lighter, human touch: Prof. Aaronson's aside about Niels Bohr hating the Many-Worlds interpretation and going on to become a nuclear-weapons strategist was a good reminder that the foundational fights in quantum mechanics exist and continue to exist.

2. What Could Be Improved

These are more personal "as a learner" notes than critiques.

Three case studies — Many-Worlds, complexity-theoretic Occam's Razor, Harlow–Hayden — in one hour was a lot for me to absorb. Each felt like it could anchor its own talk; at the delivered pace it felt rushed/compressed on my end.

The Euclid-to-Turing-to-LHC opening was lovely but long for me; I wished the back half (Harlow–Hayden) had a little more of that time.

The signature numbers — 10^6 error-correction overhead, the 10^{12} Grover threshold, $\sim 10^{500}$ Shor steps against only 10^{80} atoms in the visible universe, ~ 2^(10^{67}) ops to decode Hawking radiation, 10^{-43} s before the CPU collapses into a black hole, ~$200B in Bitcoin Shor would break — flew past me verbally and stayed scattered in my notes. One closing slide pulling them together would have helped me land on the "conservation law for hardness" thesis rather than the individual case studies.

3. Conclusions / Key Takeaways

- **Delightful delivery.** Prof. Aaronson's humor (the Bohr aside, the black-hole CPU joke about overclocking past Planck time) didn't feel like decoration — it carried me through the harder technical moments. Something I'd like to borrow the next time I'm explaining dense material.
- **Complexity as physical explanation.** The core thesis — that computational complexity can do genuine explanatory work in physics, not just bound engineering — seems increasingly mainstream from my limited vantage. The Harlow–Hayden argument, where a paradox is dissolved because the required computation is *physically impossible on complexity grounds*, was the strongest example I heard in the hour.
- **Quantum mechanics is probability theory with minus signs.** This framing is the clearest on-ramp I've come across for why quantum mechanics computes what it does. I'd love to see it used more widely in intro material.
- **Hype vs. reality in quantum computing.** Grover is √N, not exponential. Error correction eats roughly six orders of magnitude of overhead. Shor is a real problem for public-key crypto (and for the ~$200B of elliptic-curve-secured Bitcoin), but most other "quantum ML" wins are matched classically via dequantization. Thresholds I want to keep in mind the next time I hear a quantum-advantage claim.
- **Deutsch's resource question as a curious way into Many-Worlds.** Rather than arguing interpretations on aesthetic grounds, asking "where are the 10^{500} computational

steps actually occurring?" is one of the more interesting angles on Many-Worlds I've seen. It forces each interpretation to pay its conceptual bill in the same currency.

- **A possible conservation law for hardness.** Every attempted dodge around a complexity wall — faster clocks, relativistic frames, more space, more qubits — seems to migrate the cost into another conserved resource. Leaves me curious whether physics and complexity are entangled more deeply than either field typically surfaces.

Overall, one of the most rewarding hours I've spent on this material. The ideas — quantum mechanics as probability with minus signs, Deutsch's resource question, complexity as physical explanation, a possible conservation law for hardness — are the kind I expect will keep paying off for me long after the hour ends.